Boundary-sensitive products and periodic carry patterns in a two-dimensional radix system

David Gaden

12 October 2026

Abstract

We study finite integer arrays on a two-dimensional lattice under the carry move that removes ten units at one site and adds one at each of its two forward neighbours. A fixed signed digit convention gives a finite canonical representative. Oppositely directed truncated fractional cascades retain exact endpoint identities. For a product of $a \geq 1$ cascades in one direction and $c \geq 1$ in the other, at a common cutoff, we give a finite necessary-and-sufficient test for its eventual canonical display period. The least period divides 10^{a+c-1} and is computable without enumerating a full cycle. Limiting displays alone do not support multiplication. As a geometric application, an injective two-channel encoding admits exact dot-product and determinant tests with fixed nonzero array-valued factors. Literal integer endpoint encoding loses directional information, whereas retained directional units preserve it.

Draft status. This manuscript assembles current proofs and reproducible experiments for specialist review. Priority and novelty have not been established by a complete literature review. No physical model, performance advantage, or complete arithmetic of arbitrary infinite real inputs is claimed. The large phase-collision example remains in a separate companion certificate and is not needed for the main theorem here.

1 Context and elementary arithmetic

The author developed the underlying idea around 2000 and explored it with a C++ numerical application at that time. A new implementation followed in 2025. The present work formalizes patterns observed in these experiments, particularly the interaction of oppositely directed fractional cascades.

Multi-dimensional numbers (MDNs), called 2DNs in two dimensions, arrange integer digits on a lattice. The entire spreadsheet-like grid represents one number; its cells are the digits of that number. Positional order increases rightward and upward, with degree $i + j$ at site (i, j). The horizontal convention is reversed from an ordinary written numeral. This positional order does not assign an ordinary scalar weight to either axis. Arithmetic is specified by the carry rule.

A finite grid has finitely many nonzero cells. Two finite grids represent the same MDN when a finite sequence of carries connects them, allowing negative as well as positive carry amounts. A carry of $k \in \mathbb{Z}$ at (i, j) makes the three changes

$$d_{i,j} \mapsto d_{i,j} - 10k, \qquad d_{i+1,j} \mapsto d_{i+1,j} + k, \qquad d_{i,j+1} \mapsto d_{i,j+1} + k.$$

Addition is coefficientwise, and multiplication is finite convolution. They are associative, commutative and distributive modulo carries.

Laurent polynomials record positions, with X and Y denoting shifts. For general integer base $b \geq 3$, the finite carry algebra is

$$\mathcal{A}_b = \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}]/(X + Y - b).$$

Here $X + Y = b$ is exactly the local carry, not an assignment of ordinary scalar magnitudes to either axis. Eliminating the formal symbol Y gives

$$\mathcal{A}_b \cong \mathbb{Z}[X, X^{-1}, (b - X)^{-1}] \subset \mathbb{Q}(X).$$

This algebraic embedding proves that the finite algebra has no zero divisors and that $1 \neq 0$.

Digit systems in quotient rings have an established literature [1]. The polynomial $10-X-Y$ is a sandpile polynomial under Strasser's definition, and his local toppling move is the positive carry above [2]. Signed digits, full finite boundaries, and multiplication of directional cascade products require the specific treatment given here. The finite-difference methods used below also belong to the established study of modular binomial sequences [3]. These connections locate the problem; they do not establish priority for the results of this draft.

1.1 Polymorphism, signs, symmetry, and changing base

Different grids can represent one carry class, a phenomenon termed *polymorphism* in the MDN interface. For example, $1 = -9 + X + Y$ in base ten. Consequently the sign of an individual digit is representation-dependent. Additive negation of a complete number is well-defined, but the carry rule does not specify a total order.

In fact no total additive order on $\mathcal{A}_{10}$ can be invariant under axis exchange $\sigma(X) = Y$, $\sigma(Y) = X$. The nonzero element $X - Y$ would have to have the same sign as its negative $Y - X$. Total orders obtained by additional choices are not excluded by this argument.

Integer input at the origin is invariant under σ, and so is its canonical display defined below. The converse is false: XY is symmetric but is not an ordinary integer. The unity line $i + j = 0$ is different from the reflection axis $i = j$. Selecting one fractional direction can break axis-exchange symmetry; multiplying opposite directions can restore it. Thus being below the unity line does not itself imply asymmetry.

Base is part of the carry relation. Keeping the same two shift labels while changing b cannot preserve both carries and the embedded integers: the same $X + Y$ would equal two different integers. This does not imply that every encoding between bases loses information. For example, after extending coefficients to $\mathbb{Q}$, the substitution $X_b \mapsto (b/c)X_c$, $Y_b \mapsto (b/c)Y_c$ gives an isomorphism between the base-b and base-c algebras. It need not preserve the subset of finite integer-coefficient grids. A claim about base conversion therefore needs an explicit conversion map and coefficient domain.

2 A canonical finite display

Fix $\mathcal{D} = \{-4, -3, \ldots, 5\}$ and define

$$h(v) = \left\lfloor \frac{v + 4}{10} \right\rfloor, \qquad r(v) = v - 10h(v) \in \mathcal{D}.$$

Sweep through diagonals of increasing $i+j$, leaving $r(v)$ and forwarding $h(v)$ to both neighbours.

Proposition 2.1. *Every finite grid has exactly one finite representative with digits in $\mathcal{D}$, and the sweep produces it.*

Proof. Above the highest original input diagonal, if incoming coefficients have absolute value at most M, the next layer has bound $2\lfloor (M+5)/10 \rfloor$. This is strictly smaller than M for $M \geq 2$; for $M \leq 1$ no carry occurs. Each layer has finite support, so the sweep terminates. For uniqueness, a finite zero-equivalent grid is $(X+Y-10)H$ with finite H. Its lowest nonzero diagonal has all coefficients divisible by ten. The difference of two $\mathcal{D}$-grids has every coefficient between -9 and 9, so a nonzero such difference cannot be zero-equivalent. □

This normalization is a comparison convention. It is not an assertion that the GUI's Positive sign convention is canonical.

3 Directional cascades and retained boundaries

Use positive term count n; the GUI's zero-based depth is $N = n-1$. Define finite grids

$$L_n = \sum_{i=0}^{n-1} (-1)^i X^{-i-1} Y^i, \qquad R_n = \sum_{i=0}^{n-1} (-1)^i X^i Y^{-i-1}.$$

They are the finite directional entries labelled $0.1X$ and $0.1Y$. Set $q = -Y/X$ and $E_n = q^n$. Telescoping gives

$$10L_n = 1 - E_n, \qquad 10R_n = 1 - E_n^{-1}, \qquad 10^{a+c} L_n^a R_n^c = (1-E_n)^a (1-E_n^{-1})^c. \tag{1}$$

All endpoints are retained. In particular

$$100 L_n R_n = 2 - E_n - E_n^{-1}, \qquad 10 L_n R_n = L_n + R_n. \tag{2}$$

The second identity follows from the first and torsion-freeness.

One possible rigorous infinite object is an eventual equivalence class of sequences of complete finite grids: two sequences agree when their full carry classes agree at every sufficiently late stage. Addition and multiplication are stagewise and well-defined. This records an infinite recipe; it does not assert that visible limiting arrays support those operations. In such a recipe $E_n E_n^{-1} = 1$ at every stage, so identifying both endpoints with zero is incompatible with multiplication.

For independent cutoffs the exact identity instead is

$$10^{a+c} \prod_i L_{n_i} \prod_j R_{m_j} = \prod_i (1-q^{n_i}) \prod_j (1-q^{-m_j}).$$

The period theorem below concerns common cutoffs only.

4 An exact algorithm for all matched-product periods

Let $a, c \geq 1$, $d = a + c$, $r = d - 1$, and $G_n = L_n^a R_n^c$. A shift $s > 0$ is an *eventual display period* if the canonical displays of G_{n+s} and G_n agree in each fixed finite window for all sufficiently large n. The threshold may depend on the window.

Before carrying, the coefficient at $(k-a, -k-c)$ is $(-1)^k C_n(k)$, with

$$C_n(k) = [z^{k+a(n-1)}](1 + z + \cdots + z^{n-1})^d.$$

Expansion as $(1-z^n)^d/(1-z)^d$ gives, for fixed k and sufficiently large n, $C_n(k) = P(n,k) + J(k)$, where

$$P(n,k) = \sum_{u=0}^{a-1} (-1)^u \binom{d}{u} \binom{(a-u)n + k + c - 1}{r}$$

and $J(k)$ is independent of n. Specifically, $J(k) = 0$ for $k < a$ and $J(k) = (-1)^a \binom{d}{a}\binom{k+c-1}{r}$ otherwise. Binomials with arbitrary integer upper arguments are interpreted as falling-product polynomials.

For $0 \le j < r$, define

$$Q_{j,s}(n,k) = \sum_{u=0}^{a-1} (-1)^u \binom{d}{u} \left[\binom{(a-u)(n+s)+k+c-1}{r-j} - \binom{(a-u)n+k+c-1}{r-j} \right]. \quad (3)$$

Theorem 4.1. *A positive integer s is an eventual display period precisely when*

$$10^{j+1} \mid Q_{j,s}(n,k) \quad (0 \le j < r,\ n,k \ge 0,\ n+k \le r-j-1). \quad (4)$$

The minimal period $T_{a,c}$ divides 10^r. Starting with 10^r and successively removing factors of 2 and 5 whenever (4) passes gives exactly $T_{a,c}$.

Proof. Index layer j by $(k-a, -k-c+j)$. Incoming coefficients obey $v_0(k) = (-1)^k C_n(k)$ and $v_{j+1}(k) = h(v_j(k)) + h(v_j(k-1))$. Equal incoming residues modulo ten give equal digits. Conditional on agreement of previous layers, the incoming difference for a cutoff shift s is

$$\frac{(-1)^k}{10^j} \nabla_k^j \{P(n+s,k) - P(n,k)\}, \qquad \nabla f(k) = f(k) - f(k-1).$$

Forward differences equal backward differences after translating k, and the jth forward difference is (3). The initial difference has degree at most $r-1$ in k, so its rth difference vanishes. Thus agreement of all digits is equivalent to divisibility by 10^{j+1} for every integer n, k and $0 \le j < r$.

For completeness this equivalence also holds with eventual agreement. Every fixed digit depends on only finitely many raw ancestors, putting them in the polynomial interior for sufficiently large n. Conversely, an integer-valued polynomial of degree at most t is periodic modulo M in each argument: a shift by $M \operatorname{lcm}(1,\dots,t)$ preserves its binomial-basis expansion. Eventual divisibility therefore implies divisibility at every integer argument. Constant polynomials are immediate.

The total degree of $Q_{j,s}$ in (n,k) is at most $t = r-j-1$. Its Newton expansion has the form

$$Q(n,k) = \sum_{i+h \le t} (\Delta_n^i \Delta_k^h Q)(0,0) \binom{n}{i}\binom{k}{h}.$$

All coefficients are integer combinations of values in the finite triangle $n, k \ge 0$, $n+k \le t$. Divisibility there is therefore necessary and sufficient everywhere, proving (4).

Set $U = 10^r$. In the binomial translation formula for each summand of (3), the coefficients include $\binom{vU}{h}$, where $v = a-u \ge 1$ and $1 \le h \le r-j$. For $p = 2, 5$,

$$v_p\binom{vU}{h} \ge r - v_p(h) \ge r - (h-1) \ge j+1.$$

Every summand is divisible by 10^{j+1}, so U is a period. Integer periods form a subgroup of $\mathbb{Z}$ (negative shifts interpreted eventually), so the least positive period divides U. A failed prime division means its current prime exponent is minimal; removing the other prime cannot change this. The stated reduction algorithm is exact. □

The finite test has at most $r(r+1)(r+2)/6$ sample congruences per candidate. Its $n = 0$ nodes evaluate the interior polynomial; they do not describe a physical zero-term cascade. If either factor count is zero, all fixed raw coefficients stabilize, and carry locality gives period 1.

Corollary 4.2. *Writing $d = a + c$, the integer*

$$T_0 = \frac{10^{d-1}}{\gcd(10^{d-1}, \binom{d-2}{a-1})}$$

divides $T_{a,c}$. In particular $T_{1,c} = 10^c$ and $T_{a,1} = 10^a$.

Proof. The highest comparison difference is $Q_{r-1,s} = (-1)^{a-1}\binom{d-2}{a-1}s$ by the alternating binomial sum. Its required divisibility gives the lower bound. For $a = 1$ or $c = 1$ it coincides with the upper bound. □

The short lower bound need not be the period. For $(a, c) = (2, 8)$ it gives 125,000,000, but at $j = 6, n = 1, k = 0$ the remainder in (4) is 5,000,000 modulo 10^7. The exact algorithm gives 250,000,000.

(a,c)	(1,1)	(2,1)	(2,2)	(3,1)	(3,2)	(4,2)	(3,3)
$T_{a,c}$	10	100	500	1,000	10,000	25,000	50,000

5 Phases, hidden information, and a 10-adic coordinate

The theorem produces a limiting array along any sequence of growing term counts in a fixed period residue. These deterministic arrays are the "fuzzy phases": unrestricted growth need not select one of them. Cycle length does not automatically count distinct arrays.

At a fixed residue label n, equality of phases at n and $n + s$ is equivalent to

$$10^{j+1} \mid Q_{j,s}(n, k), \qquad 0 \le j < r, \quad 0 \le k \le r - j - 1.$$

This is the same proof with a one-variable Newton test in k.

Proposition 5.1. *Equality of limiting canonical arrays is not sufficient to define multiplication consistently with these finite products.*

Proof. Take term counts $n_t = 100t + 1$ and $m_t = n_t + 10$. The L_nR_n arrays have the same limits, by period 10. The L_n arrays also have the same limit, by one-direction stabilization. Their products $L_n^2R_n$ have different limits: the highest comparison difference in Corollary 4.2 requires a phase-label difference divisible by 100, whereas $m_t - n_t = 10$. Thus equal limiting inputs give unequal limiting products. □

Compatible residues modulo $10, 10^2, \ldots$ define a 10-adic integer $\alpha \in \mathbb{Z}_{10} = \varprojlim \mathbb{Z}/10^k\mathbb{Z}$. Theorem 4.1 shows that α determines the joint phase data for every matched product: a product of degree d needs at most its residue modulo 10^{d-1}. Conversely the probes $L_nR_n^c$ distinguish every residue modulo 10^c, since their highest comparison difference is simply s. Thus the joint phase family distinguishes every such α. This is a coordinate on the joint observations, not an identification of the complete boundary-retaining algebra with $\mathbb{Z}_{10}$.

6 Vector encoding and a boundary-valued circle identity

Fix $n \ge 1$ and abbreviate $L = L_n$, $R = R_n$, $F = LR$, $E = q^n$. The coefficient identities below may be read in $\mathcal{A}_{10} \otimes \mathbb{Q}$; integer coefficient inputs use only the original finite-grid algebra. The three elements $1, L, R$ are linearly independent over $\mathbb{Q}$: multiplying $A + BL + CR = 0$ by ten gives $(10A + B + C) - BE - CE^{-1} = 0$, forcing all coefficients to vanish. Here q and hence E are transcendental formal elements.

Consequently $C(a,b) = aL + bR$ is an injective additive encoding of integer vectors. It extends to rational vectors in the rational coefficient extension. It is not a ring embedding of coordinatewise vector arithmetic. The integer input parser gives the same element 1 for literal $1X$ and $1Y$, losing the distinction between $(1,0)$ and $(0,1)$. Unlike the polar origin, these are different nonzero vectors.

Direction-preserving units instead are

$$U = 10L = 1 - E, \qquad V = 10R = 1 - E^{-1}.$$

They are distinct and satisfy $UV = U + V$. No unital homomorphism to a nonzero ring can send both to 1, since it would imply $1 = 2$.

For an exact unit-circle point $(s,t) = (3/5, 4/5)$ the native decimal inputs are $6L$ and $8R$. The two proposed circle expressions are

$$(6L)(6R) + (8L)(8R) = 100F = 2 - E - E^{-1} \neq 1,$$

$$100\big((6L)^2 + (8R)^2\big) = 36(1-E)^2 + 64(1-E^{-1})^2 \neq 100.$$

For any rational s, t with $s^2 + t^2 = 1$, the consistent linear directional encoding gives the useful constant identity

$$(sU)(sV) + (tU)(tV) = UV.$$

The constant retains both endpoints. This extends to formal scalars with $s^2 + t^2 = 1$; using arbitrary real trigonometric scalars is an additional coefficient extension, not a proved property of arbitrary decimal parsing.

Proposition 6.1 (Dot product and oriented area)**.** *Let $C(a,b) = aL + bR$, let $SC(a,b) = aL - bR$ on this encoded vector space, and let σ exchange axes. Then*

$$C(a,b)\sigma C(c,d) + SC(a,b)\sigma SC(c,d) = 2(ac + bd)F,$$
$$C(a,b)\sigma C(c,d) - \sigma C(a,b)C(c,d) = (ad - bc)(L^2 - R^2).$$

The first expression vanishes exactly for orthogonal vectors; the second vanishes exactly for linearly dependent vectors.

Proof. Expand by ordinary distributivity. In the first sum the L^2 and R^2 terms cancel; in the second difference the LR terms cancel. The factors F and $L^2 - R^2$ are nonzero in the domain, so the zero tests are exact. □

Thus selected combinations of products implement geometric tests while retaining every boundary. The full product always has all four distributive terms. The operation S is defined on the vector subspace; it is not asserted to be a ring automorphism. Similarly a quarter-turn $JC(a,b) = C(-b,a)$ satisfies $J^2 = -I$ on that subspace. This does not identify an MDN axis with an imaginary unit or turn native multiplication into complex multiplication.

For $(a,b) = (6,8)$ and $(c,d) = (-8,6)$, the determinant expression is

$$100(L^2 - R^2) = -2E + E^2 + 2E^{-1} - E^{-2}.$$

All of this nonzero area information is at the retained endpoints. A sufficiently small central window therefore misses it completely.

7 Computational evidence and remaining work

The repository's `research/general_products` contains the exact finite-test implementation and 120 minimal-period certificates through total degree sixteen. A native probe linked the corrected MDN library and checked 989 finite cases with 14,832 exact checks and no failures. These compare full canonical grids and explicit carry certificates, including boundary identities. Separate exact local formula calculations check very large cutoff shifts; full native grids at those huge depths were not constructed.

The new `research/vector_circle` probe passed 45 cases and 4,045 exact checks, with zero failures. It tests rational unit-circle inputs, literal axis endpoints, full distributivity, and paired geometric identities at finite depths. Its generated `backend_results.json` is the authoritative record of the current test count and outcomes. Both studies use the corrected C++ library. The repository includes the backend corrections, regression tests, research scripts and compact result records. Reproduction commands are collected in `research/manuscript/README.md`; full generated carry certificates can be regenerated. A permanent archival release and identifier remain necessary before submission.

The next mathematical problems are to characterize a minimal multiplication-compatible boundary state, extend period analysis to independent cutoff paths, and determine the relation of these results to existing radix and algebraic-dynamics theorems. Exact geometric identities alone establish neither a faster numerical method nor a physical model. Independent specialist review and an explicit novelty comparison are required before a submission claim is made.

Development and assistance. The original construction, C++ application and motivating observations predate modern large language models. Recent AI assistance helped translate the author's computational descriptions and conjectures into mathematical definitions, proof drafts and reproducible checks. This assistance does not constitute independent specialist review.

References

[1] K. Scheicher, P. Surer, J. M. Thuswaldner and C. E. van de Woestijne, *Digit systems over commutative rings*, arXiv:1004.3729.

[2] G. Strasser, *Abelian sandpiles and algebraic models*, arXiv:1707.02281.

[3] L. Fiorot, R. Gilblas and A. Tonolo, *Periodic sequences, binomials modulo a prime power, and a math/music application*, Advances in Applied Mathematics 162 (2025), 102786, doi:10.1016/j.aam.2024.102786.